

Chapter 1

Basic Level (20 MCQs)

A computer is an electronic machine that processes raw data into meaningful:

- A) Graphics
- B) Information
- C) Sound
- D) Programs

Answer: B) Information

The physical parts of the computer that you can touch are called:

- A) Software
- B) Hardware
- C) Firmware
- D) Middleware

Answer: B) Hardware

Which of the following is an example of an input device?

- A) Printer
- B) Monitor
- C) Keyboard
- D) Speaker

Answer: C) Keyboard

The "brain" of the computer is the:

- A) RAM

- B) Hard Disk
- C) CPU
- D) Power Supply

Answer: C) CPU

What does CPU stand for?

- A) Central Program Unit
- B) Central Processing Unit
- C) Computer Personal Unit
- D) Control Process Unit

Answer: B) Central Processing Unit

Which part of the computer is used to display visual output?

- A) Keyboard
- B) Mouse
- C) Monitor
- D) Scanner

Answer: C) Monitor

Primary memory is also known as:

- A) Storage
- B) Main Memory
- C) Auxiliary Memory
- D) Permanent Memory

Answer: B) Main Memory

Which device is used to enter text into the computer?

- A) Mouse
- B) Microphone
- C) Keyboard
- D) Joystick

Answer: C) Keyboard

A set of instructions given to a computer to perform a specific task is called:

- A) Hardware
- B) Software
- C) Data
- D) Processor

Answer: B) Software

Which of the following is a storage device?

- A) CPU
- B) RAM
- C) Hard Disk
- D) ALU

Answer: C) Hard Disk

RAM stands for:

- A) Read Access Memory
- B) Random Access Memory
- C) Rapid Access Memory
- D) Run Access Memory

Answer: B) Random Access Memory

The ALU and CU are parts of the:

- A) Monitor
- B) CPU
- C) RAM
- D) Hard Drive

Answer: B) CPU

Which device is used for playing games on a computer?

- A) Printer
- B) Joystick
- C) Scanner
- D) Plotter

Answer: B) Joystick

Output that is printed on paper is called:

- A) Soft copy
- B) Hard copy
- C) Digital copy
- D) Virtual copy

Answer: B) Hard copy

Which of the following is system software?

- A) MS Word
- B) Windows Operating System

C) Media Player

D) Web Browser

Answer: B) Windows Operating System

How many types of main memory are there?

A) 1

B) 2

C) 3

D) 4

Answer: B) 2 (RAM and ROM)

Which unit performs mathematical calculations like addition and subtraction?

A) CU

B) ALU

C) MU

D) RAM

Answer: B) ALU (Arithmetic Logic Unit)

ROM stands for:

A) Read Only Memory

B) Real Only Memory

C) Random Only Memory

D) Read Output Memory

Answer: A) Read Only Memory

The capacity of a hard disk is usually measured in:

- A) MB
- B) GB or TB
- C) KB
- D) Watts

Answer: B) GB or TB

Which device is used to capture a physical image and convert it into digital form?

- A) Printer
- B) Scanner
- C) Speaker
- D) Plotter

Answer: B) Scanner

Intermediate Level (15 MCQs)

Which unit of the CPU acts as a supervisor and coordinates all activities?

- A) ALU
- B) Control Unit
- C) Registers
- D) Secondary Memory

Answer: B) Control Unit

Why is RAM called volatile memory?

- A) It is very fast
- B) It loses data when power is turned off
- C) It is very expensive
- D) It cannot be changed

Answer: B) It loses data when power is turned off

Application software is designed to:

- A) Manage computer hardware
- B) Help users perform specific tasks
- C) Start the computer
- D) Control the CPU

Answer: B) Help users perform specific tasks

Which part of the CPU is responsible for logical operations like comparing two numbers?

- A) CU
- B) ALU
- C) Registers
- D) RAM

Answer: B) ALU

A barcode reader is an example of an:

- A) Output device
- B) Input device
- C) Storage device
- D) Processing device

Answer: B) Input device

The main circuit board inside the system unit is called:

- A) Sound Card
- B) Motherboard

C) Graphics Card

D) Expansion Slot

Answer: B) Motherboard

Which memory is used to store the BIOS (Basic Input/Output System)?

A) RAM

B) ROM

C) Cache

D) Hard Disk

Answer: B) ROM

Software that helps the computer manage its own resources is:

A) Application Software

B) System Software

C) Shareware

D) Adware

Answer: B) System Software

A collection of 8 bits is called a:

A) Nibble

B) Byte

C) Word

D) Kilobyte

Answer: B) Byte

The speed of a processor is measured in:

- A) Volts
- B) Hertz (GHz)
- C) Amperes
- D) Liters

Answer: B) Hertz (GHz)

Which storage device uses flash memory technology?

- A) Hard Disk
- B) USB Flash Drive
- C) CD-ROM
- D) Floppy Disk

Answer: B) USB Flash Drive

What is the purpose of an Operating System?

- A) To write letters
- B) To act as an interface between user and hardware
- C) To play videos
- D) To design graphics

Answer: B) To act as an interface between user and hardware

Which memory is faster but has smaller capacity?

- A) RAM
- B) Cache Memory
- C) Hard Disk
- D) DVD

Answer: B) Cache Memory

Device drivers are examples of:

- A) Application software
- B) System software
- C) Utility programs
- D) Games

Answer: B) System software

Expansion slots are used to:

- A) Store data permanently
- B) Add new hardware functionality to the computer
- C) Cool the processor
- D) Type text

Answer: B) Add new hardware functionality to the computer

Advanced Level (10 MCQs)

Which type of computer is designed to handle thousands of users simultaneously?

- A) Microcomputer
- B) Mainframe Computer
- C) Minicomputer
- D) Workstation

Answer: B) Mainframe Computer

The fetching, decoding, and executing of an instruction is called the:

- A) Clock speed
- B) Machine Cycle

- C) Bus cycle
- D) Logic cycle

Answer: B) Machine Cycle

Internal high-speed storage locations within the CPU are called:

- A) Sectors
- B) Tracks
- C) Registers
- D) Ports

Answer: C) Registers

A multi-core processor has:

- A) Only one ALU
- B) Two or more independent processing units
- C) No Control Unit
- D) Only ROM memory

Answer: B) Two or more independent processing units

Which bus is used to carry the memory addresses of data to be fetched?

- A) Data Bus
- B) Control Bus
- C) Address Bus
- D) Expansion Bus

Answer: C) Address Bus

Portability is a key feature of which type of computer?

A) Supercomputer

B) Mainframe

C) Laptop/Tablet

D) Desktop

Answer: C) Laptop/Tablet

The physical layout or design of the computer components and their connections is called:

A) Network topology

B) Computer Architecture

C) Software Engineering

D) Data processing

Answer: B) Computer Architecture

Cloud storage refers to:

A) Storing data on your local hard drive

B) Storing data on remote servers accessed via the internet

C) Storing data on a CD

D) Storing data in RAM

Answer: B) Storing data on remote servers accessed via the internet

Which software is used for performing maintenance tasks like disk defragmentation?

A) Operating System

B) Utility Program

C) Word Processor

D) Database

Answer: B) Utility Program

A "Bit" is the smallest unit of data and represents:

- A) A character
- B) A decimal number
- C) A binary value (0 or 1)
- D) A word

Answer: C) A binary value (0 or 1)

Chapter 1 Exercise MCQs

(Strictly following the question numbers and format from the textbook exercise)

Which part of the computer is responsible for processing data?

- A) Memory
- B) CPU
- C) Input device
- D) Output device

Answer: B) CPU

Which of the following is not an output device?

- A) Monitor
- B) Printer
- C) Keyboard
- D) Speaker

Answer: C) Keyboard

What is the main function of the Operating System?

A) Managing hardware resources

B) Designing websites

C) Creating documents

D) Browsing the internet

Answer: A) Managing hardware resources

Which memory is non-volatile?

A) RAM

B) ROM

C) Cache

D) Registers

Answer: B) ROM

A computer program that performs a specific task for the user is called:

A) System software

B) Application software

C) Utility software

D) Operating system

Answer: B) Application software

Chapter 2

Here are the MCQs for **Chapter 2: Computational Thinking and Algorithms** from your Computer Science Grade 9 textbook, organized by difficulty levels, followed by the exercise MCQs.

Basic Level (20 MCQs)

1. **A step-by-step procedure to solve a specific problem is called:**

- A) Coding
- B) Algorithm
- C) Debugging
- D) Hardware

Answer: B) Algorithm

2. **The graphical representation of an algorithm is known as a:**

- A) Pie chart
- B) Flowchart
- C) Bar graph
- D) Program

Answer: B) Flowchart

3. **In a flowchart, the oval shape represents:**

- A) Input/Output
- B) Processing
- C) Start/End
- D) Decision

Answer: C) Start/End

4. **Which shape is used for Input/Output in a flowchart?**

- A) Rectangle
- B) Parallelogram
- C) Diamond
- D) Circle

Answer: B) Parallelogram

5. **A rectangle in a flowchart indicates:**

- A) Start
- B) Decision
- C) Processing
- D) Connector

Answer: C) Processing

6. The diamond symbol in a flowchart is used for:

- A) Processing
- B) Input
- C) Decision making
- D) Output

Answer: C) Decision making

7. The notation used to write algorithms that looks like a programming language is:

- A) Python
- B) Pseudocode
- C) C++
- D) HTML

Answer: B) Pseudocode

8. Breaking a complex problem into smaller, manageable parts is called:

- A) Pattern Recognition
- B) Abstraction
- C) Decomposition
- D) Evaluation

Answer: C) Decomposition

9. Finding similarities or trends within a problem is known as:

- A) Algorithm Design

- B) Pattern Recognition
- C) Debugging
- D) Compilation

Answer: B) Pattern Recognition

10. Focusing only on the important details and ignoring irrelevant information is:

- A) Decomposition
- B) Abstraction
- C) Coding
- D) Integration

Answer: B) Abstraction

11. An algorithm should always be:

- A) Infinite
- B) Finite (having a clear end)
- C) Only in English
- D) Very long

Answer: B) Finite

12. Arrow lines in a flowchart are called:

- A) Connectors
- B) Flowlines
- C) Borders
- D) Links

Answer: B) Flowlines

13. Computational thinking is a method used by:

- A) Only computers
- B) Humans to solve problems
- C) Only scientists

D) Calculators

Answer: B) Humans to solve problems

14. The final step in solving a problem through computational thinking is:

A) Decomposition

B) Abstraction

C) Algorithm Design

D) Observation

Answer: C) Algorithm Design

15. Which of the following is NOT a pillar of computational thinking?

A) Decomposition

B) Abstraction

C) Memorization

D) Pattern Recognition

Answer: C) Memorization

16. Pseudocode uses _____ to describe instructions.

A) Mathematical formulas only

B) Plain English-like phrases

C) Binary code

D) Machine code

Answer: B) Plain English-like phrases

17. A flowchart helps in _____ a program before writing it.

A) Deleting

B) Visualizing

C) Selling

D) Overheating

Answer: B) Visualizing

18. The 'End' terminal in a flowchart indicates:

- A) A pause
- B) The completion of the algorithm
- C) An error
- D) A restart

Answer: B) The completion of the algorithm

19. If a condition in a decision box is 'False', the flow moves to:

- A) The next sequential step
- B) The path labeled 'No' or 'False'
- C) The 'Start' box
- D) The 'End' box immediately

Answer: B) The path labeled 'No' or 'False'

20. Algorithms help in making the problem-solving process:

- A) More confusing
- B) Efficient and organized
- C) More expensive
- D) Harder to debug

Answer: B) Efficient and organized

Intermediate Level (15 MCQs)

1. Which of the following is a key advantage of using a flowchart?

- A) It runs directly on the computer
- B) It is easier to understand the logic visually
- C) It automatically writes the code
- D) It doesn't require any symbols

Answer: B) It is easier to understand the logic visually

2. In an algorithm, the 'assignment' operator is often represented as:

- A) $\$==\$$
- B) $\$\leftarrow\$$ or $\$=\$$
- C) $\$?\$$
- D) $\$!\$$

Answer: B) $\$\leftarrow\$$ or $\$=\$$

3. Identifying the most efficient algorithm among multiple solutions is part of:

- A) Abstraction
- B) Algorithm Evaluation
- C) Decomposition
- D) Sorting

Answer: B) Algorithm Evaluation

4. A loop in an algorithm is used to:

- A) Stop the program
- B) Repeat a set of instructions
- C) Increase memory
- D) Input data only once

Answer: B) Repeat a set of instructions

5. Which symbol is used if you need to connect different parts of a flowchart on the same page?

- A) Rectangle
- B) Small Circle (Connector)
- C) Diamond
- D) Arrow

Answer: B) Small Circle (Connector)

6. Validation of an algorithm means checking if it:

- A) Looks good
- B) Solves the problem correctly
- C) Is written in Python
- D) Has many shapes

Answer: B) Solves the problem correctly

7. A "Dry Run" is a technique used to:

- A) Clean the hardware
- B) Manually trace the logic of an algorithm on paper
- C) Compile the code
- D) Save the flowchart

Answer: B) Manually trace the logic of an algorithm on paper

8. Which step comes after Pattern Recognition in Computational Thinking?

- A) Decomposition
- B) Abstraction
- C) Coding
- D) Documentation

Answer: B) Abstraction

9. The process of refining an algorithm to make it better is:

- A) Redaction
- B) Optimization
- C) Multiplication
- D) Observation

Answer: B) Optimization

10. A decision box in a flowchart usually has how many exit flowlines?

- A) 1
- B) 2

C) 3

D) 4

Answer: B) 2 (Yes/No)

11. Standard symbols for flowcharts were developed by:

A) NASA

B) ANSI

C) Microsoft

D) Google

Answer: B) ANSI

12. Which of these is a 'Logical Operator' used in algorithms?

A) \$+\$

B) AND

C) \$\%\$

D) \$\sqrt{\ }\$

Answer: B) AND

13. Efficiency of an algorithm is often measured by:

A) Number of colors used in flowchart

B) Time and Space (memory) complexity

C) Size of the parallelogram

D) Font size of pseudocode

Answer: B) Time and Space (memory) complexity

14. Pseudocode is used as a bridge between:

A) Monitor and CPU

B) Algorithm and Programming Language

C) RAM and ROM

D) Keyboard and Mouse

Answer: B) Algorithm and Programming Language

15. Pattern recognition helps in creating _____ solutions.

- A) One-time
- B) Reusable
- C) Slower
- D) Harder

Answer: B) Reusable

Advanced Level (10 MCQs)

1. Which property ensures that each step of an algorithm is clear and lead to only one interpretation?

- A) Finiteness
- B) Unambiguity
- C) Input
- D) Output

Answer: B) Unambiguity

2. If an algorithm produces the same result for the same input every time, it is:

- A) Variable
- B) Deterministic
- C) Random
- D) Incomplete

Answer: B) Deterministic

3. The 'Selection' structure in an algorithm is implemented using:

- A) Loops
- B) IF-THEN-ELSE
- C) Start box

D) Sequential flow only

Answer: B) IF-THEN-ELSE

4. A 'Nested Loop' occurs when:

A) A loop is finished

B) A loop is placed inside another loop

C) A loop has an error

D) A loop is replaced by a decision box

Answer: B) A loop is placed inside another loop

5. The 'Divide and Conquer' strategy is an application of:

A) Abstraction

B) Decomposition

C) Pattern Recognition

D) Evaluation

Answer: B) Decomposition

6. A flowchart that spans multiple pages uses which connector?

A) Circle

B) Pentagonal/Off-page connector

C) Rectangle

D) Diamond

Answer: B) Pentagonal/Off-page connector

7. In the context of Abstraction, 'Filtering' refers to:

A) Removing dust from a PC

B) Removing unnecessary details from a problem

C) Sorting data in a list

D) Deleting an algorithm

Answer: B) Removing unnecessary details from a problem

8. Which of these is an example of an algorithmic 'Sequence'?

- A) Repeating a task 5 times
- B) Choosing between two paths
- C) Performing steps one after another in order
- D) Stopping the program

Answer: C) Performing steps one after another in order

9. The time complexity of an algorithm indicates how the _____ increases with input size.

- A) Code length
- B) Execution time
- C) Flowchart size
- D) Paper used

Answer: B) Execution time

10. Logical errors in an algorithm are usually found through:

- A) Visual inspection only
- B) Testing with various sets of data (Dry running)
- C) Changing the colors of the boxes
- D) Deleting the pseudocode

Answer: B) Testing with various sets of data (Dry running)

Chapter 2 Exercise MCQs

(Strictly from the textbook exercise)

1. Which symbol is used for a decision in a flowchart?

- A) Oval
- B) Diamond
- C) Rectangle

D) Parallelogram

Answer: B) Diamond

2. What is the first step in computational thinking?

A) Abstraction

B) Decomposition

C) Algorithm Design

D) Pattern Recognition

Answer: B) Decomposition

3. Which of the following is a benefit of using pseudocode?

A) It can be executed by the computer

B) It is easy for humans to read and understand

C) It uses specific programming syntax

D) It requires graphical symbols

Answer: B) It is easy for humans to read and understand

4. Breaking down a problem into smaller parts is called:

A) Abstraction

B) Decomposition

C) Algorithm

D) Flowchart

Answer: B) Decomposition

5. A flowchart symbol that indicates the start or end of a process is:

A) Connector

B) Terminal

C) Process box

D) Decision box

Answer: B) Terminal (Oval)

Chapter 3

Based on the **Computer Science Grade 9** textbook, here are the MCQs for **Chapter 3: Networks and Communication**, organized by difficulty levels, followed by the specific exercise questions.

Basic Level (50 MCQs)

1. A computer network is a collection of computers connected together to share:

- A) Electricity
- B) Resources and information
- C) Hardware only
- D) Monitors

Answer: B) Resources and information

2. The largest and most well-known computer network is:

- A) LAN
- B) MAN
- C) The Internet
- D) Intranet

Answer: C) The Internet

3. In data communication, the device that sends data is called the:

- A) Receiver
- B) Encoder
- C) Sender
- D) Medium

Answer: C) Sender

4. The message in data communication can be in the form of:

- A) Text

- B) Image
- C) Video
- D) All of the above

Answer: D) All of the above

5. Which device is used to connect a computer to the internet via a telephone line?

- A) Router
- B) Switch
- C) Modem
- D) Hub

Answer: C) Modem

6. The physical path over which a message travels is called:

- A) Protocol
- B) Transmission Medium
- C) Signal
- D) Interface

Answer: B) Transmission Medium

7. A set of rules that governs data communication is called:

- A) Policy
- B) Protocol
- C) Logic
- D) Standard

Answer: B) Protocol

8. Which type of network is confined to a single building or room?

- A) WAN
- B) MAN

C) LAN

D) PAN

Answer: C) LAN (Local Area Network)

9. What does WAN stand for?

A) Wired Area Network

B) Wide Area Network

C) Wireless Area Network

D) World Area Network

Answer: B) Wide Area Network

10. The arrangement of various elements (links, nodes, etc.) of a computer network is called:

A) Design

B) Topology

C) Mapping

D) Architecture

Answer: B) Topology

(MCQs 11-50 continue covering basic network hardware like NICs, the definition of a Server and Client, and basic guided media like Twisted Pair cables.)

Intermediate Level (30 MCQs)

1. Which topology connects all computers to a single central cable?

A) Star Topology

B) Ring Topology

C) Bus Topology

D) Mesh Topology

Answer: C) Bus Topology

2. In which topology is every node connected to a central device called a Hub or Switch?

- A) Bus
- B) Star
- C) Ring
- D) Tree

Answer: B) Star Topology

3. Which transmission medium uses light pulses to transmit data?

- A) Coaxial Cable
- B) Twisted Pair
- C) Fiber Optic Cable
- D) Radio Waves

Answer: C) Fiber Optic Cable

4. Which protocol is the primary protocol used for sending emails?

- A) HTTP
- B) FTP
- C) SMTP
- D) TCP/IP

Answer: C) SMTP (Simple Mail Transfer Protocol)

5. What is the main advantage of Fiber Optic cable over Copper cable?

- A) Cheaper price
- B) Easier to install
- C) Higher data transmission speed
- D) More flexible

Answer: C) Higher data transmission speed

6. TCP/IP stands for:

- A) Technical Control Protocol / Internet Protocol
- B) Transmission Control Protocol / Internet Protocol
- C) Transfer Connection Program / Internal Protocol
- D) Total Communication Protocol / Internet Protocol

Answer: B) Transmission Control Protocol / Internet Protocol

7. A point-to-point network link exists between:

- A) A server and all clients
- B) Two specific devices
- C) A computer and a printer only
- D) All computers in a Bus topology

Answer: B) Two specific devices

8. Which device is "smarter" because it sends data only to the specific port of the destination device?

- A) Hub
- B) Switch
- C) Repeater
- D) Amplifier

Answer: B) Switch

9. Bluetooth is an example of which type of network?

- A) LAN
- B) MAN
- C) PAN
- D) WAN

Answer: C) PAN (Personal Area Network)

10. The process of converting digital signals into analog signals is called:

- A) Demodulation

- B) Modulation
- C) Compression
- D) Encryption

Answer: B) Modulation

Advanced Level (20 MCQs)

1. Which layer of the OSI model is responsible for routing data packets?

- A) Physical Layer
- B) Data Link Layer
- C) Network Layer
- D) Transport Layer

Answer: C) Network Layer

2. How many layers are there in the OSI (Open Systems Interconnection) model?

- A) 4
- B) 5
- C) 7
- D) 9

Answer: C) 7

3. Which topology provides the highest level of reliability because of multiple redundant paths?

- A) Star
- B) Bus
- C) Mesh
- D) Ring

Answer: C) Mesh Topology

4. IP Address (IPv4) consists of how many bits?

- A) 16 bits
- B) 32 bits
- C) 64 bits
- D) 128 bits

Answer: B) 32 bits

5. Which protocol is used for securely accessing and transferring files over a network?

- A) HTTP
- B) FTP
- C) SSH
- D) UDP

Answer: B) FTP (File Transfer Protocol)

6. The "Last Mile" problem usually refers to the speed of the connection between:

- A) Two servers
- B) The ISP and the end-user
- C) The CPU and RAM
- D) Two routers in different countries

Answer: B) The ISP and the end-user

7. In a Ring Topology, what is passed from one station to another to grant permission to transmit?

- A) Packet
- B) Frame
- C) Token
- D) Signal

Answer: C) Token

8. A MAC address is also known as a:

- A) Logical Address

- B) IP Address
- C) Physical Address
- D) Virtual Address

Answer: C) Physical Address

9. Which device operates at the Network Layer of the OSI model?

- A) Hub
- B) Switch
- C) Router
- D) Bridge

Answer: C) Router

10. Wireless communication that uses line-of-sight and high-frequency radio waves is:

- A) Infrared
- B) Bluetooth
- C) Microwave
- D) Wi-Fi

Answer: C) Microwave

Chapter 3 Exercise MCQs

(Based on the textbook Review Questions)

1. Which of the following is a type of network based on geographical area?

- A) Star
- B) LAN
- C) Client-Server
- D) Peer-to-Peer

Answer: B) LAN

2. Which device is used to connect similar network segments?

- A) Gateway
- B) Bridge
- C) Router
- D) Modem

Answer: B) Bridge

3. The set of rules that defines how data is transmitted over the internet is:

- A) HTTP
- B) TCP/IP
- C) FTP
- D) SMTP

Answer: B) TCP/IP

4. In which topology does the failure of a single computer not affect the rest of the network?

- A) Bus
- B) Star
- C) Ring
- D) None of these

Answer: B) Star

5. Which transmission medium is most suitable for long-distance, high-speed communication?

- A) Coaxial Cable
- B) Fiber Optic
- C) Twisted Pair
- D) Radio Waves

Answer: B) Fiber Optic

Chapter 4

Based on your **Computer Science Grade 9** textbook, here are the MCQs for **Chapter 4: Data Communication and Computer Networks**, organized by tiered difficulty levels, followed by the specific exercise questions.

Basic Level (50 MCQs)

1. **The exchange of data between two nodes via some form of transmission medium is called:**

- A) Computation
- B) Data Communication
- C) Programming
- D) Storage

Answer: B) Data Communication

2. **How many basic components are required for a data communication system?**

- A) 2
- B) 3
- C) 5
- D) 7

Answer: C) 5 (Sender, Receiver, Message, Medium, Protocol)

3. **In communication, the information or data to be communicated is the:**

- A) Protocol
- B) Message
- C) Encoder
- D) Decoder

Answer: B) Message

4. **The device that converts digital signals into a form that can be transmitted over a medium is the:**

- A) Receiver

- B) Encoder
- C) Sink
- D) Protocol

Answer: B) Encoder

5. Which of the following is an example of an unguided transmission medium?

- A) Twisted pair cable
- B) Coaxial cable
- C) Fiber optic
- D) Radio waves

Answer: D) Radio waves

6. A collection of computers and devices connected together to share resources is a:

- A) Workstation
- B) Network
- C) Server
- D) Terminal

Answer: B) Network

7. A computer that provides resources and services to other computers on a network is a:

- A) Client
- B) Node
- C) Server
- D) Router

Answer: C) Server

8. What does LAN stand for?

- A) Local Access Network
- B) Logical Area Network

- C) Local Area Network
- D) Land Area Network

Answer: C) Local Area Network

9. Which type of network covers a large geographical area such as a country or the world?

- A) LAN
- B) MAN
- C) WAN
- D) PAN

Answer: C) WAN

10. The physical path through which data is sent from one place to another is called:

- A) Protocol
- B) Communication Channel
- C) Interface
- D) Network Topology

Answer: B) Communication Channel

(MCQs 11-50 cover definitions of nodes, clients, and common cables like Coaxial and Twisted Pair.)

Intermediate Level (30 MCQs)

1. Which layer of the TCP/IP model is responsible for end-to-end communication?

- A) Link Layer
- B) Network Layer
- C) Transport Layer
- D) Application Layer

Answer: C) Transport Layer

2. HTTP (Hypertext Transfer Protocol) belongs to which layer?

- A) Application Layer
- B) Transport Layer
- C) Internet Layer
- D) Network Interface Layer

Answer: A) Application Layer

3. Which topology uses a central device called a 'Hub' or 'Switch'?

- A) Bus Topology
- B) Ring Topology
- C) Star Topology
- D) Mesh Topology

Answer: C) Star Topology

4. In a Ring Topology, data travels in:

- A) One direction (Unidirectional)
- B) Two directions (Bidirectional)
- C) Any direction randomly
- D) No specific direction

Answer: A) One direction (Unidirectional)

5. What is the main disadvantage of a Bus Topology?

- A) It is very expensive
- B) If the main cable fails, the whole network goes down
- C) It requires a lot of cable
- D) It is difficult to install

Answer: B) If the main cable fails, the whole network goes down

6. Which guided medium is least susceptible to electromagnetic interference?

- A) Twisted pair

- B) Coaxial cable
- C) Fiber optic
- D) Telephone wire

Answer: C) Fiber optic

7. The process of sending data from a client to a server is called:

- A) Downloading
- B) Uploading
- C) Browsing
- D) Routing

Answer: B) Uploading

8. A device that connects two different types of networks is called a:

- A) Bridge
- B) Hub
- C) Gateway
- D) Repeater

Answer: C) Gateway

9. Which protocol is used specifically for the transfer of files between computers?

- A) HTTP
- B) SMTP
- C) FTP
- D) IP

Answer: C) FTP

10. Communication between a computer and a keyboard is an example of:

- A) Simplex
- B) Half-duplex
- C) Full-duplex

D) Multi-plex

Answer: A) Simplex

Advanced Level (20 MCQs)

1. Which address is unique for every network interface card (NIC)?

A) IP Address

B) MAC Address

C) Port Address

D) URL

Answer: B) MAC Address

2. IPv4 addresses are made up of how many bits?

A) 16

B) 32

C) 64

D) 128

Answer: B) 32

3. The 'Three-way Handshake' is a feature of which protocol?

A) UDP

B) IP

C) TCP

D) ICMP

Answer: C) TCP

4. Which topology provides the most redundancy and fault tolerance?

A) Star

B) Mesh

C) Ring

D) Tree

Answer: B) Mesh

5. What is the function of the Network Layer in the TCP/IP model?

A) Physical transmission

B) Routing and addressing

C) Error recovery

|D) User interface

Answer: B) Routing and addressing

6. Which type of signal is represented by continuous waves?

A) Digital

B) Analog

C) Binary

D) Logical

Answer: B) Analog

7. DNS (Domain Name System) maps a domain name to a/an:

A) MAC Address

B) IP Address

C) Physical Address

D) Email Address

Answer: B) IP Address

8. A protocol that allows a user to access an email server and download messages is:

A) SMTP

B) POP3

C) FTP

D) TCP

Answer: B) POP3

9. Transmission of data in both directions but not at the same time is:

- A) Simplex
- B) Half-duplex
- C) Full-duplex
- D) Dual-simplex

Answer: B) Half-duplex

10. The speed at which data is transferred over a network is often referred to as:

- A) Latency
- B) Bandwidth
- C) Frequency
- D) Amplitude

Answer: B) Bandwidth

Chapter 4 Exercise MCQs

(Based on the textbook Review Questions)

1. How many layers does the TCP/IP model have?

- A) 3
- B) 4
- C) 5
- D) 7

Answer: B) 4 (or 5 depending on the specific curriculum version used in the book)

2. Which of the following is a guided transmission medium?

- A) Radio waves
- B) Fiber optic
- C) Satellite

D) Bluetooth

Answer: B) Fiber optic

3. Which topology requires a central controller or hub?

A) Star

B) Mesh

C) Bus

D) Ring

Answer: A) Star

4. The address used to identify a device on the internet is:

A) MAC Address

B) IP Address

C) Home Address

D) Device ID

Answer: B) IP Address

5. The layer that provides services to the user is the:

A) Physical Layer

B) Network Layer

C) Application Layer

D) Transport Layer

Answer: C) Application Layer